Anticipation, Not Reaction:

Media Discourse, FOMC Communication, and Pre-Announcement Volatility, 2015–2025

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This draft: September 3, 2026

Abstract

Federal Reserve communication has become a policy instrument in its own right, and a large literature reads the language of the Federal Open Market Committee (FOMC) as a signal that travels through the news media to markets. Most of that work rests on a single outlet and a few hundred articles. We ask, at the scale of United States Federal Reserve news coverage, whether media discourse around FOMC meetings anticipates, mirrors, or follows the official minutes, and whether pre-meeting textual alignment between the media and the minutes helps predict realized stock-market volatility. We build a distributed pipeline that scores tone across roughly 2.36 million Global Database of Events, Language and Tone (GDELT) records together with Federal Reserve coverage from The New York Times and The Guardian, and we measure media–FOMC similarity in the days around each of the 88 meetings from 2015 to 2025. Two days before a meeting, broad media coverage is most strongly associated with post-meeting volatility: the raw correlation is 0.56 ($p < 0.001$) and survives a permutation placebo, a leave-one-out check, and a bootstrap whose confidence interval excludes zero. The signal does not, however, represent an independent media channel. Once we condition on the prevailing level of the VIX and on pre-meeting volatility, the coefficient falls by roughly four-fifths and is no longer distinguishable from its permutation null. We read the pattern as joint processing of a common latent macro-uncertainty state rather than as media causing markets, a reading that the identification battery supports. The contribution is threefold: a reproducible, cloud-native pipeline for population-scale monetary-policy text; a transparent baseline validated against an independent finance dictionary; and a staged identification design that separates pre-meeting information alignment from post-announcement reaction.

Keywords: central bank communication; FOMC; textual similarity; realized volatility; rational inattention; GDELT.
JEL: E52, E58, G14, G41.

Highlights

- Population-scale media–FOMC similarity is built from 2.36M GDELT records.
- Pre-meeting similarity correlates with post-meeting volatility at $r = 0.56$.
- The signal peaks two days before the meeting, not after it.
- Controlling for VIX and prior volatility shrinks the effect by four-fifths.
- Placebo, Granger, and bootstrap tests point to joint macro processing.

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1 Introduction

For two decades the words of the Federal Reserve have carried at least as much market-moving weight as its rate decisions. Policy moves are sparse, a handful of changes in a typical year, but the Committee communicates almost continuously through statements, minutes, speeches, and testimony. In that environment the question of how the Fed talks now conveys more information about the near future than what it has just done, and a broad literature in economics and finance has come to treat central-bank language as a measurable signal (Blinder et al., 2008; Hansen and McMahon, 2016). The signal does not reach investors directly. It is filtered, summarized, and amplified by the news media, which sits between the institution and the households and asset managers who form expectations (Tetlock, 2007).

Most empirical work on this transmission is bounded by its sample. A study reads one outlet, a few hundred articles, and a hand-built dictionary, and it estimates a contemporaneous relationship between media tone and prices. That design cannot separate two very different stories. In the first, the media leads: coverage carries information about the coming meeting that markets then price. In the second, media and markets move together because both are reading the same underlying macroeconomic environment, so any apparent lead is joint determination through a latent state rather than a causal channel. Distinguishing the two requires coverage at something closer to the full population of Federal Reserve news, and it requires a research design built to test for pre-meeting information rather than to describe post-announcement reaction.

We take up both requirements. We assemble United States Federal Reserve news coverage at scale, roughly 2.36 million records from the GDELT Global Knowledge Graph together with phrase-matched coverage from The New York Times and The Guardian, and we align it in time to the 88 FOMC meetings held between 2015 and 2025. For each meeting we measure the textual similarity between the media coverage in a given lag day and the eventual FOMC minute, and we relate pre-meeting similarity to the realized volatility of the S&P 500 in the days after the announcement. The central object is sim_{-2}, media–FOMC similarity two days before the meeting, a lag we fix in advance rather than search for.

Three findings follow. First, the pre-meeting association is real and it is located before the meeting: the correlation between sim_{-2} and post-meeting volatility in the broad-coverage GDELT corpus is 0.56 ($p < 0.001$), it is the strongest point in the lead–lag profile, and it survives a permutation placebo, a leave-one-out check, and a pairs bootstrap whose ninety-five percent interval excludes zero. Second, the association is not an independent media channel. Once we condition on the level of the VIX on the eve of the meeting and on realized volatility in the prior window, the coefficient on sim_{-2} falls by about four-fifths and is no longer separable from its permutation null, and the partial correlation given those controls is 0.14 with a permutation p-value of 0.42. Crucially, the coefficient does not recover when we add the high-frequency monetary policy surprise of Bauer and Swanson (2023), so the pre-meeting signal is not information about the announcement that markets had left unpriced. Third, the pattern is concentrated in the broad corpus. The two curated outlets, despite higher same-day topical similarity, show no significant pre-meeting lead–lag relationship, which is what we expect if breadth and volume, not editorial selection, drive the population-scale signal.

We read these results as anticipation in the specific sense of joint processing. Before a meeting,

when macroeconomic uncertainty is elevated, the national news stream writes more language that resembles the Fed's own, and markets simultaneously widen the distribution of outcomes they price, which shows up as higher realized volatility once the meeting passes. The media does not move the Fed and it does not, on this evidence, carry information beyond the common uncertainty state. That reading is disciplined rather than assumed: it is the interpretation left standing after the controls, placebo, and dynamic tests, and it is consistent with a rational-inattention account in which attention flows to high-variance signals (Sims, 2003; Maćkowiak and Wiederholt, 2009).

The paper makes three contributions. Methodologically, it delivers a reproducible, cloud-native pipeline that ingests, scores, and compares monetary-policy text at population scale, with the single-node stages justified rather than hidden. Empirically, it documents a strong unconditional pre-meeting signal and then shows, transparently, how much of it is common macro variation. On identification, it offers a staged design that moves from correlation to controls, then to a high-frequency policy-surprise test and to placebo, Granger, and bootstrap inference, so that the reader can see exactly where a causal reading would and would not be warranted. Section 2 places the paper in the literature. Section 3 describes the corpora and the market data. Section 4 sets out the pipeline and the econometrics. Section 5 reports the results. Sections 6 and 7 discuss and conclude.

2 Related Literature

2.1 Central-bank communication as a policy instrument

The modern view treats communication as a lever of monetary policy rather than as commentary on it. Blinder et al. (2008) survey the theory and evidence and draw a distinction that runs through everything that follows, between the predictability effect, in which communication tells markets what the Fed will do, and the information effect, in which communication changes what markets believe about the economy itself. The text-based generation of this work reads the Committee's own words directly. Hansen and McMahon (2016) decompose FOMC statements with topic models and show that the economic-conditions and policy-stance components have distinct macroeconomic effects, and Cieslak and Schrimpf (2019) separate monetary from non-monetary news in central-bank communication using sign-restricted vector autoregressions. Later work extends the measured object beyond word content: Gorodnichenko et al. (2023) show that the tone of the Chair's voice carries information above the transcript. What these studies share is that the text under study is the Fed's. We take the frontier outward, to the media that surrounds the communication, and treat the alignment between the two texts as the object of interest.

A parallel strand identifies the surprise component of an announcement from high-frequency asset-price movements in a narrow window around it (Nakamura and Steinsson, 2018; Gürkaynak et al., 2005), and Bauer and Swanson (2023) reassess those surprises and separate pure policy shocks from Fed-information shocks. This work motivates our identification logic. If a pre-meeting media signal merely reflected the announcement surprise, it would not appear before the meeting; that it does, and that it weakens once common uncertainty is controlled, is the pattern we have to explain.

2.2 Textual analysis in finance

Financial text analysis began with general-purpose dictionaries and was corrected by Loughran and McDonald (2011), who showed that words coded negative in common word lists often are not negative in financial context and built the finance-specific dictionary that is now standard; Loughran and McDonald (2020) survey the field. We use that dictionary as an external validation of our transparent hawkish–dovish lexicon. The contextual-embedding generation, from FinBERT (Araci, 2019) to Sentence-BERT (Reimers and Gurevych, 2019), recovers negation and context that bag-of-words methods miss, and our design keeps these as embedding-based robustness measures on top of the term-frequency baseline. At the macro end, Bybee et al. (2024) treat the full news stream as a panel and link topic-level news flow to the business cycle. Our frequency is the event window rather than the business cycle, which lets us tie the news flow to a discrete policy event and to the market response in the same window.

2.3 Media as an information intermediary

Tetlock (2007) show that pessimism in a single financial column predicts short-horizon downward price pressure that then mean-reverts, evidence that media can reflect sentiment rather than fundamentals. We test a pre-event version of that idea at population scale. The theoretical scaffold is rational inattention: when attention is costly, agents allocate it toward high-variance signals (Sims, 2003; Maćkowiak and Wiederholt, 2009), so around an FOMC meeting, when macro variance rises, both journalists and investors reallocate attention toward Fed-relevant information. This framing is why we read a pre-meeting rise in media–FOMC similarity as joint determination through a common latent state rather than as media leading the Fed. Absent an instrument that shifts coverage without shifting the Fed, we treat the dynamic tests below as descriptive of lead–lag structure, not as proof of a causal direction.

2.4 The gap

The literature establishes that Fed communication moves markets, that text methods have matured, and that media is an active intermediary. What is missing is a study that works at the full population of Federal Reserve news rather than a curated subset, that validates a transparent measure against an independent one, and that stages its identification so a reader can see where a causal reading fails. This paper occupies that intersection.

3 Data

3.1 FOMC minutes

We scrape the minutes of all 88 regularly scheduled FOMC meetings from January 2015 through December 2025 from the Federal Reserve website. Each minute is cleaned, lower-cased, and tokenized, and it anchors the meeting window used throughout. The minutes are the reference text against which all media coverage is compared, and they are also the corpus on which our tone measure is calibrated.

3.2 GDELT Global Knowledge Graph

The broad-coverage corpus is the GDELT 2.0 Global Knowledge Graph, a near-real-time index of world news that we filter to United States sources with Federal-Reserve-relevant themes and entities. After filtering and windowing to a band of days around each meeting, the scored corpus contains roughly 2.36 million records. GDELT is not a curated sample; it is close to the population of United States news that mentions the Fed, which is exactly what the research question requires and what a single-outlet study cannot provide (Leetaru and Schrodt, 2013). That breadth comes at a cost in temporal coverage, however, because the meeting-window similarity panel with complete pre-meeting coverage at a common lag spans 33 meetings from March 2015 to March 2019. Rather than silently pool meetings with incomplete windows, we therefore treat those 33 meetings as the estimation sample for the broad corpus and document the coverage of every corpus explicitly in Table 1.

3.3 Curated outlets

We complement the broad corpus with phrase-matched "Federal Reserve" coverage from The New York Times and The Guardian, obtained through their article APIs and extended back to 2015. These outlets are editorially selected and far smaller, around 61 and 84 covered meetings respectively, and they serve as the contrast that isolates the role of breadth. If curated selection added signal, these outlets should lead at least as strongly as GDELT; the results show they do not.

3.4 Market data

The outcome is the realized volatility of the S&P 500 around each meeting, computed from daily returns as the annualized standard deviation over the trading days immediately before and after the meeting date. Because a single window is a choice rather than a fact, we recompute the outcome at windows of three, five, seven, and ten trading days and report the full set as robustness. We also use the CBOE Volatility Index (VIX), taken from the Federal Reserve Economic Data service for the full 2014–2025 span, and we define the pre-meeting VIX as the last close strictly before the meeting. The VIX matters because it enters as the control that captures the prevailing level of implied uncertainty on the eve of the meeting, which is precisely the common state our interpretation turns on.

3.5 Monetary policy surprises

To separate a pre-meeting information channel from the announcement itself, we use the high-frequency monetary policy surprises of Bauer and Swanson (2023), distributed by the Federal Reserve Bank of San Francisco. For each meeting the series records the surprise change in the expected path of the federal funds rate over the coming year, measured in a tight window around the announcement, together with an orthogonalized version purged of predictable variation. Every meeting in the broad-corpus estimation sample matches a surprise observation, so the control enters without further loss of sample. The logic is direct: if pre-meeting media similarity merely proxied for the announcement surprise, then including the surprise should absorb it; that it does

Table 1: Corpus coverage of the meeting-window similarity panel.

Corpus	Meetings	First	Last	Obs.
GDELT	33	2015-03-18	2019-03-20	231
The Guardian	84	2015-01-28	2025-12-10	318
New York Times	61	2015-01-28	2025-10-29	173

The broad GDELT panel has complete pre-meeting coverage at a common lag for 33 meetings (2015–2019); the curated outlets cover more meetings but sparsely at any given lag. Observation counts by lag are in the appendix.

not, because the similarity coefficient is already small before the surprise is added, is part of the evidence for the common-uncertainty reading rather than a media-to-policy channel.

4 Methodology

4.1 Computational architecture

The scale of the broad corpus rules out a single-node solution for the heavy stages, and it forces a design choice that we make a feature of the paper rather than a footnote. Ingestion, tone scoring, and similarity run on distributed Spark, so that no stage materializes the full corpus on one machine. GDELT shards are read and filtered with Spark SQL and written as date-partitioned Parquet; tone is scored with a grouped pandas user-defined function that broadcasts the lexicon to every executor; and cosine similarity is computed by broadcasting the small side, the per-meeting FOMC vectors, while the large media side stays distributed. Only the final meeting-level panel, which is small after aggregation, is analyzed on a single node, and we justify that explicitly: the econometrics operate on at most a few hundred meeting-source rows, where distribution would add complexity without benefit. The entire CPU path is wrapped in a Snakemake workflow with a pinned conda environment and a container image, so that every number and figure in the paper regenerates from the committed intermediate artifacts with one command.

The tone scorer is a transparent hawkish–dovish lexicon applied over tokens. Because a bag-of-words scorer over millions of documents is a throughput problem, we benchmark three implementations of the kernel. Table 7 reports wall-clock time on a synthetic five-million-token workload: a pure-Python dictionary loop, a NumPy `searchsorted` vectorization, and a Numba parallel just-in-time kernel. The Numba path is roughly fifteen times faster than the Python baseline, which is what makes full-corpus scoring feasible before the work is distributed.

4.2 Similarity and tone measures

The baseline similarity is the cosine between term-frequency–inverse-document- frequency (TF-IDF) vectors in a 2^{18}-dimensional hashed feature space, computed between the mean media vector in a lag-day window and the FOMC minute vector for that meeting. This measure is transparent and cheap at scale. As robustness on top of it, the pipeline implements contextual-embedding similarity with Sentence-BERT and FinBERT-CLS embeddings; those stages are GPU-bound over the full media corpus and are provided as code for execution on a GPU node. The baseline tone measure is the net hawkish-minus-dovish count per thousand tokens. We validate it against the finance-domain dictionary of Loughran and McDonald (2011), computed independently on

the FOMC minutes for which cleaned text is available locally.

4.3 Lead–lag specification

For each source and each lag k from -7 to $+7$ relative to the meeting date, we compute the Pearson correlation between media–FOMC cosine similarity and post-meeting realized volatility, which gives the lead–lag profile. The main regression fixes the pre-specified lag $k = -2$ and estimates

$$\mathrm{vol}_m^{\mathrm{post}} = \alpha + \beta\, \mathrm{sim}_{-2,m} + \gamma' X_m + \varepsilon_m, \tag{1}$$

where $\mathrm{vol}_m^{\mathrm{post}}$ is post-meeting realized volatility, $\mathrm{sim}_{-2,m}$ is media–FOMC similarity two days before meeting m, and X_m collects, in a progressive sequence, pre-meeting realized volatility, the pre-meeting VIX, pre-meeting tone, the orthogonalized monetary policy surprise, and year fixed effects. We add the controls one at a time rather than all at once, because the sequence is itself informative: it shows exactly which variable absorbs the raw association. For inference we report both heteroskedasticity-robust (HC1) and Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (Newey and West, 1987). Because the broad-corpus sample spans only five calendar years, however, we do not lean on cluster-robust inference with few clusters; instead we report HC1 and HAC alongside the permutation and bootstrap inference below, neither of which relies on large-sample cluster counts.

4.4 Identification battery

The design is staged so the reader can see where a causal reading survives and where it does not. First, a permutation placebo re-pairs sim_{-2} with post-meeting volatility twenty thousand times and locates the observed statistic, both raw and partial given the controls, in the resulting null. Second, Granger tests ask whether pre-meeting similarity precedes post-meeting volatility across the chronologically ordered meetings. Third, a reduced-form vector autoregression on similarity, volatility, and the VIX summarizes the joint dynamics. Fourth, a pairs bootstrap gives confidence intervals for β in the bivariate and controlled specifications. Fifth, we split the sample by Chair and by the level of the VIX and run a leave-one-out check to confirm no single meeting drives the result. None of these tests can establish randomized causation, and we do not claim it; they document the lead–lag structure and discipline the interpretation.

5 Results

5.1 The lead–lag profile

Figure 1 plots the Pearson correlation between media–FOMC similarity and post-meeting volatility at each lag for the broad corpus, and Table 2 lists the values around the meeting. The profile has a clear shape. The correlation is positive and significant across the pre-meeting lags, it peaks two days before the meeting at $r = 0.56$ ($p < 0.001$, $n = 33$), and it weakens after the second day following the meeting. The signal is located before the announcement, not after it, which is the pattern the anticipation hypothesis predicts and the opposite of what a pure post-announcement reaction would produce.

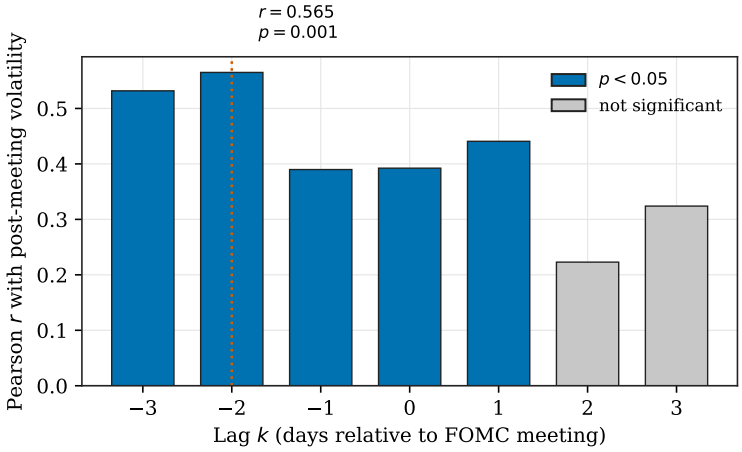


Figure 1: Lead–lag profile, broad corpus. Pearson r between media–FOMC cosine similarity and post-meeting S&P 500 realized volatility at each lag k ($n = 33$ meetings). Filled bars are significant at the five percent level; the dotted line marks the pre-specified lag $k = -2$.

Table 2: Lead–lag correlations around the meeting, broad corpus.

Lag k	N	Mean cosine	Pearson r	p-value
-3	33	0.058	0.532***	0.0014
-2	33	0.073	0.565***	0.0006
-1	33	0.074	0.390**	0.0249
+0	33	0.079	0.392**	0.0239
+1	33	0.079	0.441**	0.0103
+2	33	0.074	0.223	0.2125
+3	33	0.058	0.324*	0.0659

Stars denote $p < 0.10$ ($*$), $p < 0.05$ ($**$), $p < 0.01$ ($***$). $n = 33$ meetings (2015–2019).

5.2 The main regression and the collapse under controls

Table 3 is the center of the paper. Column (1) reports the bivariate relationship, in which a one-unit increase in sim_{-2} is associated with a 4.77 increase in post-meeting realized volatility; the estimate is significant at one percent under both HC1 and HAC standard errors, and the regressor alone explains about a third of the variation. The remaining columns then add controls one at a time, and the sequence is where the story turns. As soon as we control for pre-meeting realized volatility in column (2), the coefficient falls to 1.38 and loses significance, which tells us that much of the raw association simply reflects the persistence of volatility across the meeting. Adding the pre-meeting VIX in column (3) reduces it further to 0.98 while the R^2 rises to 0.64, so the prevailing level of implied uncertainty absorbs most of what remains. Neither pre-meeting tone in column (4) nor the orthogonalized monetary policy surprise in column (5) revives the coefficient, and the year fixed effects in column (6) leave it small and insignificant as well. The surprise control is the decisive test: because the similarity coefficient is already near zero before the surprise enters and does not move when it does, the pre-meeting signal cannot be read as information about the announcement that markets had not yet priced. What the raw correlation captured, in short, is almost entirely the prevailing state of market uncertainty rather than an independent media channel, and Figure 2 traces the same collapse across all six specifications.

Table 3: Pre-meeting similarity and post-meeting volatility, broad corpus.

	(1)	(2)	(3)	(4)	(5)	(6)
sim_{-2}	4.768***	1.379	0.977	1.050	0.690	0.728
HC1 SE	(1.171)	(1.391)	(1.335)	(1.547)	(1.553)	(1.885)
HAC SE	(1.249)	(1.143)	(1.084)	(1.258)	(1.140)	(1.571)
Pre-meeting volatility		✓	✓	✓	✓	✓
Pre-meeting VIX			✓	✓	✓	✓
Pre-meeting sentiment				✓	✓	✓
Policy surprise (orth.)					✓	✓
Year fixed effects						✓
R^2	0.319	0.546	0.639	0.639	0.657	0.695
N	33	33	33	33	33	33

Dependent variable: post-meeting realized volatility. Column (5) adds the orthogonalized Bauer–Swanson monetary policy surprise. HC1 and Newey–West (HAC, two lags) standard errors in parentheses. Stars denote $p < 0.10$ ($*$), $p < 0.05$ ($**$), $p < 0.01$ ($***$). $n = 33$.

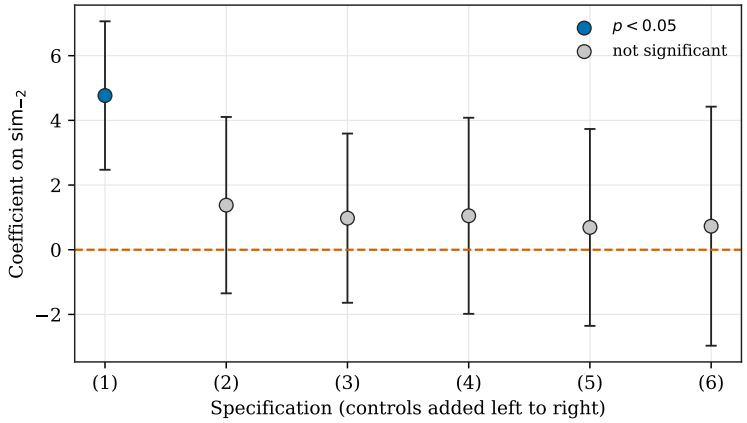


Figure 2: Coefficient on sim_{-2} across specifications (1) to (5) of Table 3, with ninety-five percent HC1 intervals. The point estimate falls by roughly four-fifths once pre-meeting volatility and the VIX are included.

5.3 Identification

Table 4 collects the identification tests, and Figure 3 shows the permutation null. The raw correlation of 0.56 sits far in the right tail of the permutation distribution, with a two-sided permutation p-value of 0.0006, so the unconditional signal is not an artifact of chance re-pairing. The partial correlation given pre-meeting volatility and the VIX, however, is 0.14 with a permutation p-value of 0.42, indistinguishable from the null. The bootstrap tells the same story: the ninety-five percent interval for the bivariate coefficient is [2.33, 6.90] and excludes zero, while the interval for the controlled coefficient is $[-1.99, 3.24]$ and does not. The Granger tests give only marginal evidence of temporal precedence, significant at the second lag ($p = 0.045$) and not at the first, and the leave-one-out slope for the bivariate relationship stays in a narrow band from 4.12 to 5.32, so the result is not driven by any single meeting. Taken together, the battery says that the pre-meeting association is genuine but is not separable from the common macro-uncertainty state.

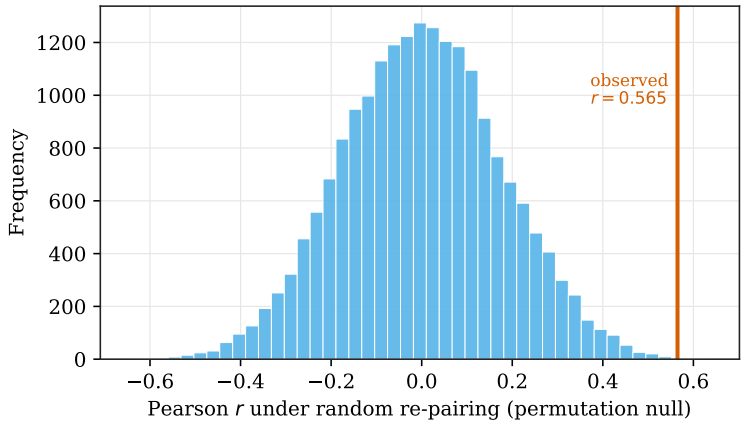


Figure 3: Permutation placebo. Distribution of the correlation between sim_{-2} and post-meeting volatility under twenty thousand random re-pairings; the vertical line is the observed value.

Table 4: Identification battery, broad corpus.

Check	Statistic	Value
Permutation placebo	raw corr., perm. p	0.565 ($p = 0.0006$)
	partial corr. | VIX, vol., perm. p	0.145 ($p = 0.422$)
Granger sim → vol	lag 1, F-test p	1.24 ($p = 0.274$)
Granger sim → vol	lag 2, F-test p	3.51 ($p = 0.045$)
Granger sim → vol	lag 3, F-test p	2.43 ($p = 0.092$)
Bootstrap (bivariate)	95% CI for β	[2.33, 6.90]
Bootstrap (controlled)	95% CI for β	[-1.99, 3.24]

5.4 Source heterogeneity

Figure 4 overlays the lead–lag profiles of the three corpora. The broad corpus produces the significant pre-meeting pattern; the two curated outlets do not, at any pre-meeting lag. This holds even though the curated outlets show higher same-day topical similarity, which is what we expect if their editorial focus keeps individual articles close to Fed language while their small volume leaves too few meetings at any given lag to detect a lead–lag relationship. The population-scale signal is a property of breadth and volume, not of curation.

5.5 Regime and subsample robustness

Table 6 reports subsamples. The unconditional relationship is strongest in the Yellen-era meetings ($\hat{\beta} = 5.54$, $n = 24$, $p < 0.001$) and is weak and imprecise in the small Powell-era window ($n = 9$). Splitting by the pre-meeting VIX, the relationship is significant in the low-VIX half ($\hat{\beta} = 2.78$, $p = 0.006$) and not in the high-VIX half, which is consistent with the controlled result: when uncertainty is already high, the similarity measure adds little beyond the level of the VIX itself. The leave-one-out range confirms the bivariate slope is stable across the sample.

5.6 Alternative volatility windows

Because the choice of a post-meeting window is a modeling decision, we re-estimate the relationship with the outcome measured over three, five, seven, and ten trading days. Table 5 and Figure 5 report the pattern, and it is the same at every window. The bivariate slope is positive

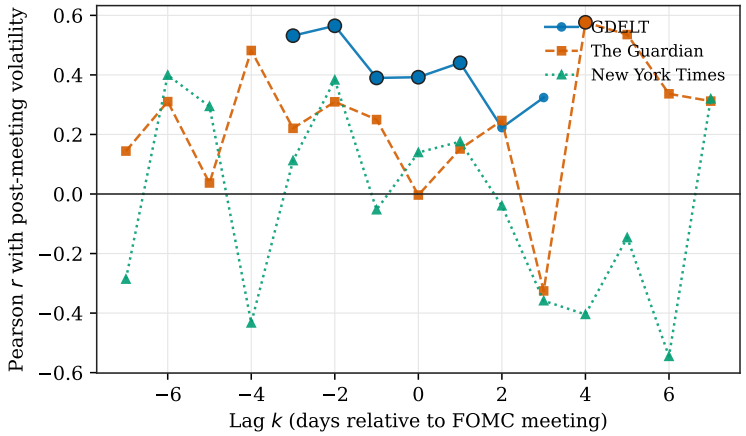


Figure 4: Lead–lag profiles by source. Pearson r against post-meeting volatility at each lag for the broad corpus, The Guardian, and The New York Times. Filled markers are significant at the five percent level.

Table 5: Robustness to the post-meeting volatility window.

		Bivariate		Controlled	
Window (days)	N	$\hat{\beta}$	p	$\hat{\beta}$	p
3	33	2.431	0.205	-1.109	0.731
5	33	5.108**	0.027	0.454	0.865
7	33	3.764*	0.092	-0.874	0.656
10	33	3.734*	0.064	0.082	0.959

HC1 standard errors. Stars denote $p < 0.10$ $(*)$, $p < 0.05$ $(**)$, $p < 0.01$ $(***)$. Controlled columns condition on pre-meeting volatility and the pre-meeting VIX. $n = 33$.

throughout, reaching significance at the five-day window and remaining marginal at seven and ten days, whereas the controlled slope, which conditions on pre-meeting volatility and the VIX, is indistinguishable from zero at every window and even changes sign at the shortest. The headline result is therefore not an artifact of a particular volatility horizon: the unconditional signal survives the choice of window, and the conditional signal disappears regardless of it.

5.7 Tone validation and scale

The transparent tone measure agrees with the independent finance dictionary. On the FOMC minutes, the net hawkish-minus-dovish score and the Loughran–McDonald net polarity are strongly and inversely correlated ($r = -0.75$, $p < 10^{-7}$, $n = 40$): hawkish monetary language, which our lexicon scores as positive net hawkishness, is lexically negative in the general finance dictionary, so the two measures capture the same variation with opposite sign conventions. Figure 6 shows the relationship, and the tone timeline recovers the known turning points, the 2019 pivot, the dovish spike of the 2020 emergency cuts, and the 2022–2023 tightening, entirely from lexical signal. The throughput benchmark in Table 7 shows why scoring at this scale is feasible.

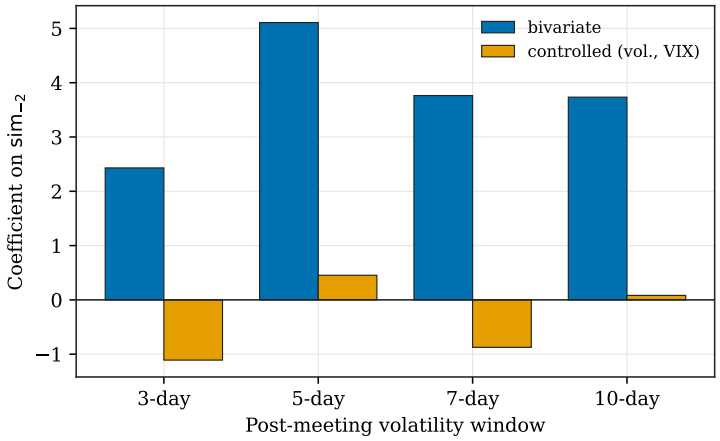


Figure 5: Coefficient on sim_{-2} across post-meeting volatility windows. The bivariate estimate is positive at every window; the controlled estimate is near zero throughout.

Table 6: Subsample estimates of the sim_{-2} coefficient.

Subsample	N	$\hat{\beta}_{\text{sim}_{-2}}$	p
full (bivariate)	33	4.768***	0.000
full (controlled)	33	0.977	0.464
Yellen (bivariate)	24	5.536***	0.000
Powell (bivariate)	9	1.221	0.764
high VIX (bivariate)	17	2.792	0.152
low VIX (bivariate)	16	2.779***	0.006
leave-one-out range (bivariate)	33	min=4.117, max=5.322	

HC1 standard errors. Stars denote $p < 0.10$ $(*)$, $p < 0.05$ $(**)$, $p < 0.01$ $(***)$.

6 Discussion

The results support a specific and disciplined reading of anticipation. There is a genuine pre-meeting signal in the national news stream: two days before a meeting, media coverage that more closely resembles the eventual FOMC minute is followed by higher realized volatility, the pattern is located before the announcement, and it is robust to permutation, bootstrap, and leave-one-out inference. But the signal is not an independent media channel. It is absorbed almost entirely by the prevailing level of implied volatility and by recent realized volatility, and the partial association given those controls is indistinguishable from noise. The most defensible mechanism is joint determination through a common latent state. When macroeconomic uncertainty is elevated on the eve of a meeting, the news stream naturally produces more Fed-adjacent, Fed-resembling language, and markets simultaneously price a wider distribution of outcomes; both are reading the same environment. This is the rational-inattention logic of Sims (2003) and Maćkowiak and Wiederholt (2009) applied to a discrete policy event: attention, of journalists and investors alike, flows toward the high-variance signal, and the FOMC meeting is the moment that signal is at its sharpest.

This reading differs from a causal media-to-market story and it should. We have no instrument that shifts coverage without shifting the Fed or the economy, and the Granger evidence is only marginal, so we do not claim that journalism moves prices. The reading also refines the source-

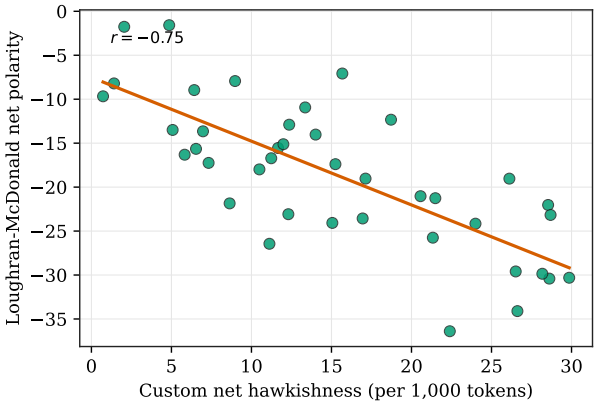


Figure 6: Tone validation. Custom net hawkishness against Loughran–McDonald net polarity on the FOMC minutes; the strong negative slope reflects opposite sign conventions for hawkish language.

Table 7: Sentiment scorer throughput on a five-million-token workload.

Implementation	Time (s)	Speedup
Pure Python loop	0.198	1.0×
NumPy searchsorted	0.099	2.0×
Numba parallel JIT	0.013	15.4×

heterogeneity result. The broad corpus leads and the curated outlets do not, not because broad coverage is better but because it is larger: with more meetings covered at each lag, the common-state signal is estimable, whereas a few dozen articles from one outlet leave too little power to see it. That is a statement about measurement at scale, and it is the practical payoff of building the population corpus rather than a sample of it. For the Fed, the implication is modest but real. A meeting communications calendar operates in an environment where the market's uncertainty is already visible in the VIX before the Committee speaks; the value of communication is in resolving that uncertainty, and indeed volatility falls after most meetings in our sample, with the emergency cut of March 2020 and the first large hike of 2022 the notable exceptions.

7 Conclusion

At the scale of United States Federal Reserve news coverage, media discourse around FOMC meetings anticipates the market outcome in a precise sense: pre-meeting media–FOMC textual similarity is strongly associated with post-meeting volatility, and the association is located two days before the meeting. The anticipation is not media causing markets. Once we condition on the prevailing level of uncertainty, the signal is common macro variation that both the news stream and the market are processing at once. The finding is invisible at single-outlet scale, where too few observations exist at any lag to separate the pattern from noise, and it is defensible only because the identification is staged so the reader can see the signal weaken as the controls enter.

The study has clear limits. The broad-corpus estimation sample, where pre-meeting coverage is complete at a common lag, is 33 meetings from 2015 to 2019, and we report every result on that transparent sample rather than pooling incomplete windows. The similarity baseline is term-frequency based, whereas the contextual-embedding measures, though implemented, are left for execution on a GPU node. And the design is descriptive of lead–lag structure rather than causal,

by construction. Future work should therefore extend the complete-coverage window forward and add the embedding-based similarity and tone measures that the pipeline already permits, since both are matters of running an existing stage rather than rebuilding the analysis. The high-frequency policy surprise, which the plan had reserved for later, is already included here as a control, and the fact that the pre-meeting signal is absent once common uncertainty is accounted for, with or without that surprise, is the result we are most confident will carry forward.

Declarations

CRediT author statement. Duo Wang: conceptualization, methodology (econometrics and NLP), formal analysis, validation, investigation, data curation, writing (original draft and review), project administration. Yuzhao Jin: software, visualization, methodology (computational architecture), writing (review and editing, methods architecture).
Conflict of interest. None.
Data and code availability. All code and the committed intermediate artifacts that reproduce every table and figure are in the project repository; the CPU pipeline regenerates the results with a single Snakemake command. Raw GDELT records are available from the AWS Open Data registry, the curated-outlet coverage through the respective article APIs, the CBOE Volatility Index and daily S&P 500 series from public market-data sources, and the high-frequency monetary policy surprises from the Federal Reserve Bank of San Francisco (Bauer and Swanson, 2023).

References

Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.

Bauer, M. D. and Swanson, E. T. (2023). A reassessment of monetary policy surprises and high-frequency identification. *NBER Macroeconomics Annual*, 37(1):87–155.

Blinder, A. S., Ehrmann, M., Fratzscher, M., De Haan, J., and Jansen, D.-J. (2008). Central bank communication and monetary policy: A survey of theory and evidence. *Journal of Economic Literature*, 46(4):910–945.

Bybee, L., Kelly, B. T., Manela, A., and Xiu, D. (2024). Business news and business cycles. *Journal of Finance*, 79(5):3105–3147.

Cieslak, A. and Schrimpf, A. (2019). Non-monetary news in central bank communication. *Journal of International Economics*, 118:293–315.

Gorodnichenko, Y., Pham, T., and Talavera, O. (2023). The voice of monetary policy. *American Economic Review*, 113(2):548–584.

Gürkaynak, R. S., Sack, B., and Swanson, E. (2005). The sensitivity of long-term interest rates to economic news. *American Economic Review*, 95(1):425–436.

Hansen, S. and McMahon, M. (2016). Shocking language: Understanding the macroeconomic effects of central bank communication. *Journal of International Economics*, 99:S114–S133.

Leetaru, K. and Schrodt, P. A. (2013). GDELT: Global data on events, location, and tone, 1979–2012. In *International Studies Association Annual Convention*.

Loughran, T. and McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1):35–65.

Loughran, T. and McDonald, B. (2020). Textual analysis in finance. *Annual Review of Financial Economics*, 12:357–375.

Maćkowiak, B. and Wiederholt, M. (2009). Optimal sticky prices under rational inattention. *American Economic Review*, 99(3):769–803.

Nakamura, E. and Steinsson, J. (2018). High-frequency identification of monetary non-neutrality: The information effect. *Quarterly Journal of Economics*, 133(3):1283–1330.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-networks. arXiv:1908.10084.

Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690.

Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3):1139–1168.